Image Classification Extension

https://github.com/aiyshwary/Image-Classification-Extension-in-Rapid-Miner

Python | PyTorch | ONNX | RapidMiner | Transfer Learning

OVERVIEW

A RapidMiner Studio extension enabling no-code transfer learning for image classification, supporting fine-tuning of popular CNN architectures on custom datasets and ONNX-based inference without writing a single line of code.

IMPACT

Developed a RapidMiner extension with two custom operators — Fine Tuning and Inference — enabling practitioners to apply transfer learning with ResNet, VGGNet, DenseNet, Inception and more using a drag-and-drop interface.

KEY DETAILS

- Fine Tuning operator accepts an image directory and hyperparameters; outputs a fine-tuned model in ONNX format.
- Inference operator loads the ONNX model and outputs a prediction DataFrame with classified labels.
- Supports ResNet, AlexNet, VGGNet, SqueezeNet, DenseNet, and Inception architectures.
- Compatible with both CPU and GPU (CUDA) environments via configurable conda environment.
- Installed as a JAR extension in RapidMiner Studio with Python Scripting backend.